

Supplementary Material for “Texture-Invariant Conformal Prediction and Detection in Compound-Gaussian Radar Clutter: Exact Laws, Thermal Noise, and Evaluation on Two Radars”

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All numbers below are generated from the saved outcomes in the project repository (`research/r7c`). Four protocols were frozen and hash-recorded before their outcomes were computed: the confirmatory IPIX study (`PROTOCOL.md`), detection (`detection/PROTOCOL_DETECTION.md`), baselines (`baselines/PROTOCOL_BASELINES.md`) and the second radar (`jku/PROTOCOL_JKU.md`).

1 Hold-out series

Two further McMaster series (#269 and #287, the widely used “high” and “low” sea-state files, one provider-preprocessed VV cell each, from the same 1993 campaign) were hashed and left unopened until the main analysis was complete, and then evaluated once (Table 1). They confirm the direction of the main result, with larger gains. They also show a case in which the classical Gaussian threshold over-covers while conformal IN-ARCP stays near nominal. Their test sets are small (about 165 episodes per unit), so the hold-out is a directional check. Mondrian calibration returns infinite regions at $1 - \alpha = 0.99$ there, because each bin receives fewer calibration episodes than the rank requires.

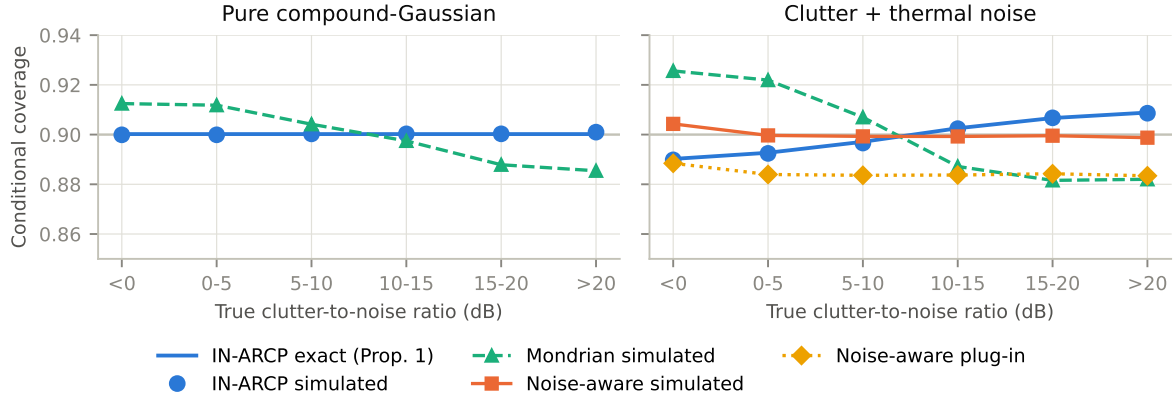
Table 1: Hold-out series #269 and #287 ($1 - \alpha = 0.90$; six units per column).

Method	$m = 8$		$m = 16$	
	Radius ratio	Coverage (min)	Radius ratio	Coverage (min)
Noise-aware AR(4) (proposed)	0.454	0.890 (0.816)	0.498	0.893 (0.847)
Noise-aware AR(2) (proposed)	0.541	0.887 (0.828)	0.533	0.895 (0.859)
AR(4), innovation-normalized	0.546	0.908 (0.847)	0.515	0.906 (0.883)
AR(2), innovation-normalized	0.582	0.905 (0.840)	0.572	0.906 (0.871)
Noise-aware AR(1)	0.921	0.898 (0.840)	0.921	0.903 (0.859)
IN-ARCP, AR(1)	1.000	0.906 (0.853)	1.000	0.910 (0.859)
Gaussian plug-in of IN-ARCP	1.178	0.949 (0.896)	1.104	0.945 (0.895)
Mondrian IN-ARCP	1.000	0.895 (0.804)	0.981	0.896 (0.810)
Learned-scale CP	0.629	0.893 (0.804)	0.662	0.919 (0.791)
Invariant MLP CP	1.809	0.884 (0.736)	2.158	0.903 (0.761)
Raw-RMS normalized CP	1.321	0.880 (0.706)	1.243	0.887 (0.706)
Unnormalized CP	1.142	0.868 (0.607)	1.161	0.859 (0.589)
Gaussian plug-in of noise-aware AR(4)	0.411	0.866 (0.816)	0.472	0.856 (0.755)

2 Synthetic verification

3 Proof sketch of Proposition 4 (zero-noise limit)

With $\gamma = \nu c$ the history log-likelihood converges locally uniformly on $\gamma > 0$ to $-m \log \gamma - \gamma^{-1} h^H R_H^{-1} h$, whose unique maximizer is $h^H R_H^{-1} h / m$; coercivity at 0 and ∞ gives convergence of the maximizers. The Markov property gives $\mu \rightarrow \rho h_m$, and $(1 - |\rho|^2) R_H^{-1} = W_\rho^H W_\rho$, with W_ρ the bidiagonal whitening matrix, identifies the limiting predictive variance with $s_\rho(h)^2$.



200 replications per panel, $m = 16$, nominal 0.90; Monte Carlo SE ≤ 0.0014 .

Figure 1: Synthetic verification: coverage conditional on the true clutter-to-noise ratio. The line is the exact law of Proposition 1; markers are simulation.

Table 2: Day transfer ($m = 16$, $1 - \alpha = 0.90$): fit and calibrate on other days, test on each of six held-out days.

Method	Mean coverage	Worst held-out day	Radius ratio
Noise-aware AR(4) (proposed)	0.898	0.863	0.936
Noise-aware AR(2) (proposed)	0.898	0.883	0.956
AR(4), innovation-normalized	0.901	0.894	0.993
AR(2), innovation-normalized	0.899	0.891	1.024
Noise-aware AR(1)	0.900	0.882	1.037
IN-ARCP, AR(1)	0.903	0.882	1.000
Gaussian plug-in of IN-ARCP	0.906	0.893	1.006
Mondrian IN-ARCP	0.907	0.881	0.984
Learned-scale CP	0.898	0.774	0.967
Invariant MLP CP	0.880	0.622	1.200
Raw-RMS normalized CP	0.906	0.681	1.466
Unnormalized CP	0.909	0.716	1.470
Gaussian plug-in of noise-aware AR(4)	0.876	0.842	0.893

4 Day transfer and texture-proxy sensitivity

Texture-proxy sensitivity (pre-specified windows; mean maximum quintile deviation at $1 - \alpha = 0.90$, $m = 16$): with ± 512 pulses IN-ARCP 0.035, NA-AR(4) 0.043, unnormalized 0.181, neural centre 0.103; with ± 2048 pulses IN-ARCP 0.033, unnormalized 0.141. The ranking is the same as with ± 1024 .

5 Additional figures for the IPIX study

6 Detection study: all levels and methods

Pooled detection probability by SCR, measured false-alarm rates, exact-law predictions and SCR gains with 95% day-cluster intervals (output of `detection/analyze_detection.py`):

units: 56; days: 6; test episodes: 74980

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=== 1-alpha = 0.9 (Swerling 1; Pd pooled mean over units; SCR in dB rel. local clutter+noise power) ===
method Pfa_meas -5.0 -2.5 0.0 2.5 5.0 7.5 10.0 12.5 15.0 17.5 20.0 22.5 25.0
CAlloc 0.1030 0.166 0.219 0.307 0.433 0.579 0.711 0.814 0.888 0.934 0.963 0.979 0.988 0.993
CA16 0.1002 0.242 0.312 0.406 0.516 0.631 0.736 0.823 0.887 0.931 0.959 0.977 0.987 0.992
U 0.1011 0.446 0.549 0.651 0.745 0.823 0.883 0.926 0.955 0.974 0.985 0.992 0.995 0.997
RMS 0.1007 0.442 0.541 0.642 0.734 0.813 0.877 0.922 0.953 0.972 0.984 0.991 0.995 0.997
MLP 0.0990 0.447 0.548 0.650 0.743 0.823 0.884 0.927 0.956 0.974 0.985 0.992 0.995 0.997
LS 0.0995 0.534 0.624 0.710 0.784 0.847 0.898 0.935 0.960 0.977 0.987 0.993 0.996 0.997
MDN 0.0961 0.540 0.634 0.720 0.794 0.858 0.907 0.941 0.965 0.979 0.988 0.993 0.996 0.998
G1 0.1004 0.542 0.635 0.722 0.795 0.859 0.908 0.942 0.965 0.980 0.989 0.993 0.996 0.998
IN1 0.0987 0.540 0.633 0.720 0.793 0.857 0.906 0.941 0.964 0.979 0.988 0.993 0.996 0.998
IN2 0.0991 0.553 0.643 0.728 0.801 0.862 0.909 0.943 0.966 0.980 0.989 0.993 0.996 0.998
IN4 0.0985 0.574 0.661 0.741 0.811 0.869 0.914 0.946 0.967 0.981 0.989 0.994 0.996 0.998
NA1 0.0994 0.553 0.648 0.735 0.810 0.872 0.917 0.948 0.969 0.982 0.990 0.994 0.997 0.998
NA2 0.0987 0.609 0.692 0.767 0.834 0.887 0.926 0.954 0.972 0.984 0.991 0.995 0.997 0.998
NA4 0.0982 0.614 0.696 0.770 0.836 0.888 0.927 0.954 0.973 0.984 0.991 0.995 0.997 0.998
NA4G 0.1150 0.630 0.710 0.782 0.845 0.895 0.932 0.958 0.975 0.986 0.992 0.995 0.997 0.998
IN1 exact-law prediction 0.507 0.614 0.710 0.791 0.856 0.906 0.942 0.965 0.979 0.988 0.993 0.996 0.998

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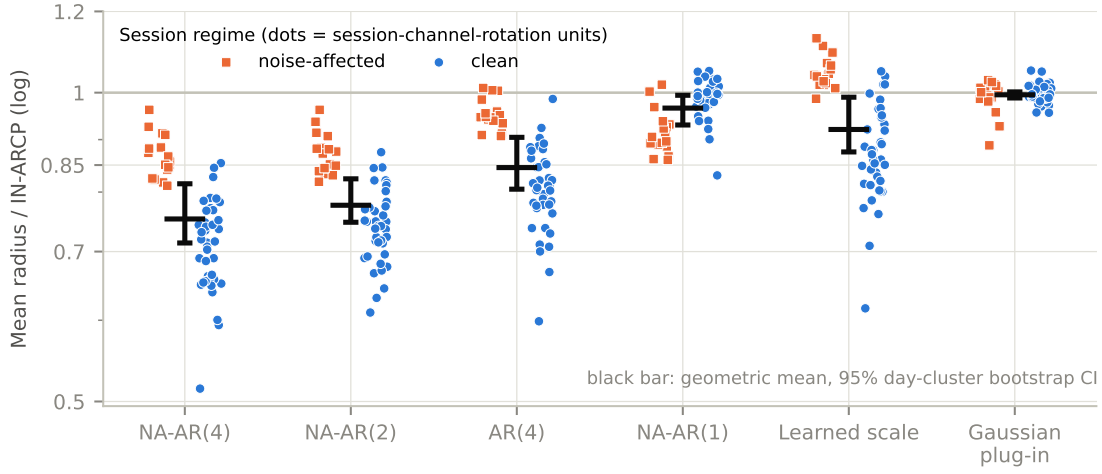


Figure 2: Per-unit radius relative to IN-ARCP at $1 - \alpha = 0.90$, $m = 16$, by session regime (noise-affected: fitted noise power above 1% of mean power).

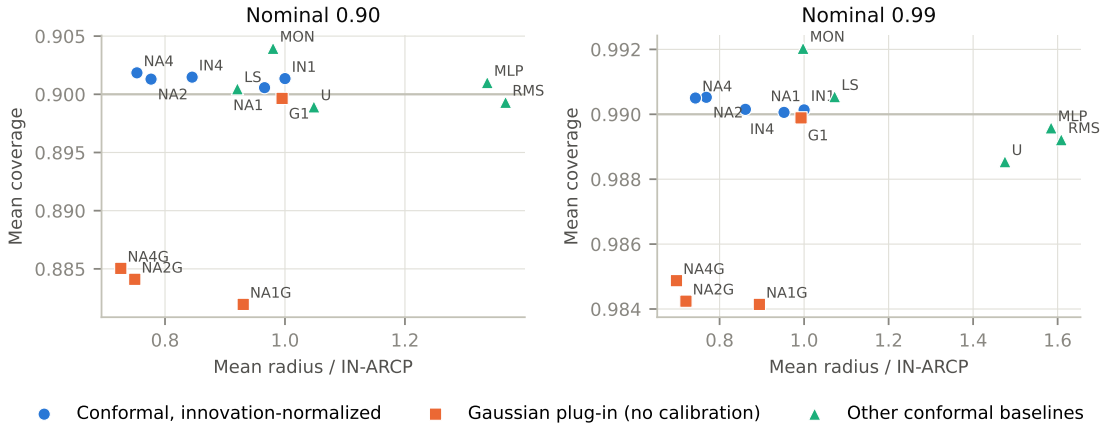


Figure 3: Mean coverage against mean radius (confirmatory, $m = 16$). Gaussian plug-ins of the richer noise-aware models are shorter but under-cover; conformal calibration restores nominal coverage. Labels: NAp and INp , noise-aware and innovation-normalized $AR(p)$ ($IN1$ is IN-ARCP); $G1$ and $NApG$, Gaussian plug-ins; MON , Mondrian; LS , learned scale; U , unnormalized; RMS , raw-RMS; MLP , neural centre.

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exact-law MAE, IN1 Pd: per unit x SCR 0.0140; pooled curve 0.0053
-- SCR gain (dB) needed for Pd=0.5; positive = less SCR than reference; 95% day-cluster CI --
IN1 SCR -3.0 dB | vs CA16 5.06 [ 3.25, 6.46] | vs CALoc 6.64 [ 4.50, 8.41]
IN4 SCR -3.1 dB | vs CA16 5.19 [ 3.41, 6.53] | vs CALoc 6.76 [ 4.69, 8.46]
NA4 SCR -3.4 dB | vs CA16 5.41 [ 3.85, 6.61] | vs CALoc 6.99 [ 5.10, 8.52]
G1 SCR -3.1 dB | vs CA16 5.10 [ 3.35, 6.44] | vs CALoc 6.67 [ 4.65, 8.39]
U SCR -2.6 dB | vs CA16 4.61 [ 2.79, 6.04] | vs CALoc 6.19 [ 4.15, 8.03]
MLP SCR -2.6 dB | vs CA16 4.66 [ 2.72, 6.11] | vs CALoc 6.23 [ 3.87, 8.10]
LS SCR -2.8 dB | vs CA16 4.83 [ 2.83, 6.34] | vs CALoc 6.40 [ 4.06, 8.30]
-- SCR gain (dB) needed for Pd=0.9; positive = less SCR than reference; 95% day-cluster CI --
IN1 SCR 5.2 dB | vs CA16 7.98 [ 5.03,10.43] | vs CALoc 7.93 [ 5.02,10.37]
IN4 SCR 4.3 dB | vs CA16 8.86 [ 5.48,11.49] | vs CALoc 8.80 [ 5.45,11.44]
NA4 SCR 3.1 dB | vs CA16 10.05 [ 6.52,12.77] | vs CALoc 9.99 [ 6.51,12.72]
G1 SCR 5.1 dB | vs CA16 8.02 [ 5.09,10.42] | vs CALoc 7.97 [ 5.11,10.38]
U SCR 7.1 dB | vs CA16 6.09 [ 3.33, 8.23] | vs CALoc 6.04 [ 3.30, 8.21]
MLP SCR 7.0 dB | vs CA16 6.12 [ 3.70, 7.88] | vs CALoc 6.07 [ 3.70, 7.81]
LS SCR 5.4 dB | vs CA16 7.78 [ 4.32,10.40] | vs CALoc 7.72 [ 4.35,10.37]

=== 1-alpha = 0.99 (Swirling 1; Pd pooled mean over units; SCR in dB rel. local clutter+noise power) ===
method Pfa_meas -5.0 -2.5 0.0 2.5 5.0 7.5 10.0 12.5 15.0 17.5 20.0 22.5 25.0
CALoc 0.0110 0.022 0.034 0.062 0.125 0.237 0.394 0.566 0.714 0.822 0.893 0.939 0.966 0.980
CA16 0.0103 0.081 0.129 0.203 0.306 0.434 0.568 0.695 0.796 0.872 0.922 0.954 0.974 0.985
U 0.0115 0.131 0.208 0.308 0.428 0.558 0.680 0.782 0.860 0.913 0.948 0.970 0.983 0.990
RMS 0.0108 0.176 0.271 0.383 0.503 0.625 0.733 0.822 0.887 0.931 0.960 0.977 0.987 0.993
MLP 0.0104 0.176 0.272 0.388 0.510 0.632 0.742 0.831 0.893 0.935 0.962 0.978 0.988 0.993
LS 0.0095 0.288 0.391 0.495 0.597 0.694 0.776 0.846 0.900 0.938 0.963 0.979 0.988 0.993
MON 0.0080 0.326 0.434 0.546 0.651 0.743 0.822 0.883 0.927 0.956 0.974 0.985 0.992 0.995
G1 0.0101 0.337 0.448 0.558 0.661 0.751 0.828 0.887 0.929 0.957 0.975 0.986 0.992 0.995
IN1 0.0099 0.335 0.445 0.554 0.657 0.747 0.825 0.885 0.927 0.956 0.975 0.986 0.992 0.995
IN2 0.0102 0.357 0.464 0.569 0.670 0.758 0.832 0.889 0.931 0.959 0.976 0.986 0.992 0.996
IN4 0.0098 0.377 0.482 0.586 0.681 0.765 0.836 0.892 0.932 0.960 0.977 0.987 0.992 0.996
NA1 0.0099 0.354 0.466 0.577 0.681 0.773 0.848 0.903 0.940 0.965 0.980 0.989 0.994 0.996
NA2 0.0095 0.423 0.531 0.632 0.723 0.802 0.868 0.915 0.948 0.969 0.982 0.990 0.994 0.997
NA4 0.0095 0.433 0.539 0.637 0.727 0.805 0.869 0.916 0.948 0.969 0.982 0.990 0.994 0.997
NA4G 0.0151 0.465 0.566 0.659 0.745 0.819 0.878 0.921 0.952 0.971 0.984 0.991 0.995 0.997
IN1 exact-law prediction 0.281 0.406 0.531 0.645 0.743 0.824 0.885 0.929 0.957 0.975 0.986 0.992 0.995

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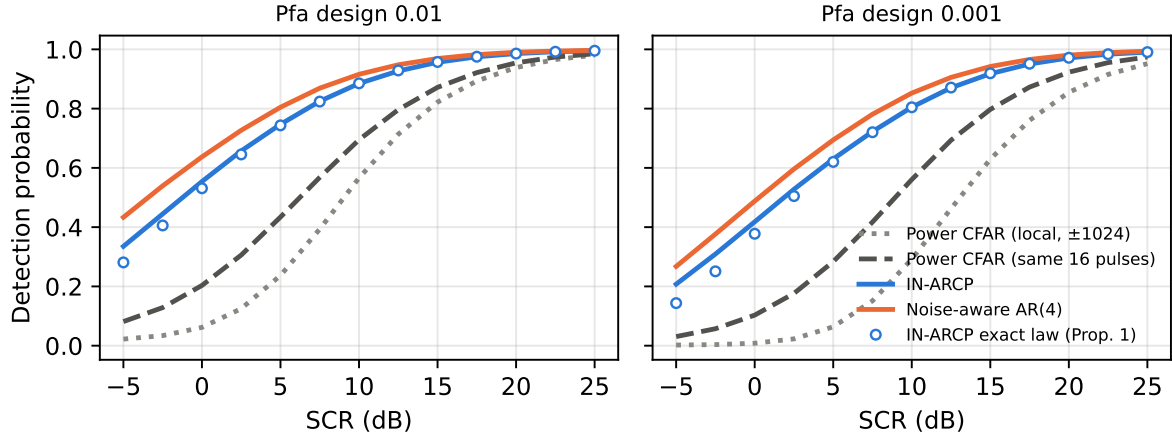


Figure 4: Detection of injected Swerling-1 targets on IPIX at design false-alarm probabilities 0.01 and 0.001.

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exact-law MAE, IN1 Pd: per unit x SCR 0.0210; pooled curve 0.0105
-- SCR gain (dB) needed for Pd=0.5; positive = less SCR than reference; 95% day-cluster CI --
IN1 SCR -0.9 dB | vs CA16 6.96 [ 4.29, 9.04] | vs Caloc 9.81 [ 6.41,12.61]
IN4 SCR -1.3 dB | vs CA16 7.37 [ 4.54, 9.52] | vs Caloc 10.22 [ 6.67,13.26]
NA4 SCR -1.9 dB | vs CA16 8.03 [ 5.45,10.02] | vs Caloc 10.88 [ 7.56,13.66]
G1 SCR -1.0 dB | vs CA16 7.05 [ 4.40, 9.11] | vs Caloc 9.90 [ 6.48,12.71]
U SCR 3.8 dB | vs CA16 2.30 [ 0.78, 3.44] | vs Caloc 5.15 [ 3.34, 6.61]
MLP SCR 2.2 dB | vs CA16 3.88 [ 1.89, 5.34] | vs Caloc 6.73 [ 3.89, 8.70]
LS SCR 0.4 dB | vs CA16 5.72 [ 2.75, 7.96] | vs Caloc 8.57 [ 4.89,11.49]
-- SCR gain (dB) needed for Pd=0.9; positive = less SCR than reference; 95% day-cluster CI --
IN1 SCR 8.7 dB | vs CA16 7.62 [ 4.55,10.01] | vs Caloc 8.97 [ 5.58,11.67]
IN4 SCR 8.0 dB | vs CA16 8.35 [ 4.95,10.96] | vs Caloc 9.70 [ 5.90,12.64]
NA4 SCR 6.5 dB | vs CA16 9.81 [ 6.27,12.45] | vs Caloc 11.16 [ 7.28,14.14]
G1 SCR 8.7 dB | vs CA16 7.67 [ 4.71,10.02] | vs Caloc 9.02 [ 5.70,11.73]
U SCR 13.5 dB | vs CA16 2.80 [ 1.01, 4.34] | vs Caloc 4.15 [ 2.23, 5.73]
MLP SCR 12.0 dB | vs CA16 4.36 [ 2.30, 5.85] | vs Caloc 5.71 [ 3.23, 7.35]
LS SCR 10.2 dB | vs CA16 6.14 [ 2.75, 8.69] | vs Caloc 7.49 [ 3.78,10.46]

=== 1-alpha = 0.999 (Swerling 1; Pd pooled mean over units; SCR in dB rel. local clutter+noise power) ===
method Pfa_meas -5.0 -2.5 0.0 2.5 5.0 7.5 10.0 12.5 15.0 17.5 20.0 22.5 25.0
CA1oc 0.0008 0.002 0.004 0.008 0.023 0.064 0.151 0.292 0.463 0.629 0.761 0.855 0.915 0.952
CA16 0.0007 0.031 0.057 0.103 0.176 0.284 0.417 0.561 0.693 0.797 0.873 0.923 0.955 0.974
U 0.0014 0.023 0.048 0.092 0.159 0.251 0.362 0.485 0.606 0.710 0.796 0.863 0.911 0.944
RMS 0.0007 0.057 0.108 0.184 0.287 0.411 0.539 0.665 0.772 0.853 0.910 0.946 0.969 0.982
MLP 0.0009 0.057 0.110 0.190 0.297 0.426 0.559 0.683 0.788 0.865 0.918 0.952 0.972 0.984
LS 0.0010 0.124 0.198 0.290 0.390 0.494 0.596 0.693 0.778 0.849 0.902 0.940 0.964 0.979
MDN 0.0000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
G1 0.0013 0.226 0.333 0.446 0.560 0.665 0.756 0.834 0.893 0.934 0.961 0.977 0.987 0.993
IN1 0.0011 0.208 0.309 0.418 0.528 0.630 0.723 0.805 0.870 0.918 0.951 0.971 0.984 0.991
IN2 0.0009 0.209 0.309 0.418 0.526 0.629 0.722 0.803 0.868 0.916 0.950 0.970 0.983 0.990
IN4 0.0008 0.230 0.332 0.440 0.543 0.641 0.729 0.804 0.866 0.913 0.947 0.969 0.982 0.990
NA1 0.0012 0.219 0.325 0.440 0.554 0.661 0.758 0.837 0.897 0.937 0.963 0.979 0.988 0.993
NA2 0.0009 0.272 0.380 0.491 0.598 0.695 0.783 0.853 0.906 0.943 0.966 0.981 0.989 0.994
NA4 0.0007 0.267 0.377 0.488 0.596 0.694 0.781 0.852 0.905 0.942 0.966 0.981 0.989 0.994
NA4G 0.0028 0.370 0.477 0.581 0.674 0.760 0.833 0.890 0.931 0.958 0.976 0.986 0.992 0.996
IN1 exact-law prediction 0.144 0.251 0.378 0.505 0.620 0.720 0.804 0.871 0.919 0.951 0.972 0.984 0.991
exact-law MAE, IN1 Pd: per unit x SCR 0.0266; pooled curve 0.0155
-- SCR gain (dB) needed for Pd=0.5; positive = less SCR than reference; 95% day-cluster CI --
IN1 SCR 1.9 dB | vs CA16 6.84 [ 4.08, 8.94] | vs Caloc 10.95 [ 7.28,13.95]
IN4 SCR 1.7 dB | vs CA16 7.07 [ 4.04, 9.44] | vs Caloc 11.19 [ 7.42,14.41]
NA4 SCR 0.5 dB | vs CA16 8.27 [ 5.76,10.32] | vs Caloc 12.39 [ 8.88,15.43]
G1 SCR 1.2 dB | vs CA16 7.52 [ 5.19, 9.39] | vs Caloc 11.64 [ 8.28,14.48]
U SCR 10.4 dB | vs CA16 -1.64 [-4.25,-0.30] | vs Caloc 2.48 [ 0.60, 3.74]
MLP SCR 6.2 dB | vs CA16 2.55 [ 1.07, 3.82] | vs Caloc 6.67 [ 3.97, 8.33]
LS SCR 5.1 dB | vs CA16 3.69 [ 1.34, 5.58] | vs Caloc 7.81 [ 4.35,10.22]
-- SCR gain (dB) needed for Pd=0.9; positive = less SCR than reference; 95% day-cluster CI --
IN1 SCR 11.5 dB | vs CA16 7.22 [ 4.16, 9.55] | vs Caloc 9.57 [ 5.97,12.67]
IN4 SCR 11.2 dB | vs CA16 7.48 [ 4.09,10.22] | vs Caloc 9.90 [ 6.03,13.47]
NA4 SCR 9.7 dB | vs CA16 8.96 [ 6.01,11.33] | vs Caloc 11.37 [ 7.84,14.47]
G1 SCR 10.8 dB | vs CA16 7.88 [ 5.23,10.04] | vs Caloc 10.22 [ 6.92,13.23]
U SCR 18.6 dB | vs CA16 0.08 [-1.16, 0.98] | vs Caloc 2.35 [ 0.65, 3.73]
MLP SCR 15.9 dB | vs CA16 2.85 [ 1.26, 4.16] | vs Caloc 5.32 [ 3.01, 6.76]
LS SCR 14.6 dB | vs CA16 4.10 [ 1.22, 6.30] | vs Caloc 6.51 [ 2.89, 9.48]

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7 Persistent targets after onset

Frozen onset study (detection/PROTOCOL_ONSET.md): per-look detection probability, cumulative detection within K looks, and clean cumulative false-alarm probabilities:

units 56; segments 74980

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=== per-look Pfa design 0.01 ===
clean cumulative false-alarm prob, K=1,2,4,8: IN1 0.0099 0.0189 0.0357 0.0683
clean cumulative false-alarm prob, K=1,2,4,8: G1 0.0101 0.0192 0.0369 0.0706
clean cumulative false-alarm prob, K=1,2,4,8: IN4 0.0098 0.0191 0.0369 0.0721
clean cumulative false-alarm prob, K=1,2,4,8: NA4 0.0095 0.0182 0.0363 0.0701
clean cumulative false-alarm prob, K=1,2,4,8: CA16 0.0103 0.0170 0.0302 0.0540
clean cumulative false-alarm prob, K=1,2,4,8: Caloc 0.0113 0.0147 0.0209 0.0308
-- per-look Pd at SCR 10 dB, Doppler random (looks j=0..7) --
IN1 0.895 0.711 0.639 0.478 0.084 0.012 0.005 0.002
G1 0.897 0.715 0.645 0.497 0.068 0.011 0.005 0.003
IN4 0.891 0.725 0.563 0.256 0.022 0.003 0.001 0.001
NA4 0.914 0.883 0.705 0.571 0.431 0.287 0.201 0.178
CA16 0.691 0.601 0.455 0.254 0.100 0.039 0.017 0.009

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CAlloc 0.562 0.564 0.564 0.563 0.565 0.565 0.563 0.565
-- per-look Pd at SCR 10 dB, Doppler matched (looks j=0..7) --
IN1 0.885 0.024 0.013 0.007 0.003 0.002 0.002 0.001
G1 0.887 0.025 0.014 0.008 0.004 0.003 0.002 0.002
IN4 0.891 0.049 0.004 0.001 0.001 0.001 0.001 0.001
NA4 0.914 0.666 0.153 0.015 0.006 0.004 0.003 0.003
CA16 0.691 0.610 0.460 0.186 0.033 0.013 0.006 0.004
CAlloc 0.562 0.563 0.564 0.564 0.564 0.565 0.565 0.565
-- per-look Pd at SCR 10 dB, Doppler opposite (looks j=0..7) --
IN1 0.885 0.940 0.905 0.796 0.165 0.014 0.004 0.002
G1 0.887 0.942 0.909 0.817 0.127 0.014 0.005 0.002
IN4 0.891 0.927 0.763 0.087 0.009 0.002 0.001 0.001
NA4 0.914 0.945 0.931 0.831 0.675 0.500 0.380 0.351
CA16 0.691 0.591 0.454 0.268 0.090 0.040 0.015 0.008
CAlloc 0.562 0.564 0.564 0.563 0.564 0.564 0.565 0.562
-- SCR (dB) for cumulative Pd=0.5 by K, Doppler random; gain vs CAlloc [95% day-cluster CI] --
IN1 K=1: 9.82 [ 6.40,12.68] | K=2: 9.92 [ 6.58,12.67] | K=4: 9.19 [ 5.99,11.87] | K=8: 8.42 [ 5.22,11.05]
NA4 K=1: 10.93 [ 7.46,13.74] | K=2: 10.34 [ 7.31,12.81] | K=4: 9.82 [ 7.17,11.99] | K=8: 9.40 [ 7.19,11.28]
IN4 K=1: 10.23 [ 6.71,13.18] | K=2: 9.88 [ 6.44,12.62] | K=4: 9.14 [ 5.82,11.82] | K=8: 8.28 [ 4.83,11.05]
G1 K=1: 9.90 [ 6.58,12.74] | K=2: 9.97 [ 6.75,12.63] | K=4: 9.27 [ 6.23,11.85] | K=8: 8.53 [ 5.56,11.06]
CA16 K=1: 2.76 [ 1.81, 4.06] | K=2: 2.94 [ 1.88, 4.41] | K=4: 2.85 [ 1.67, 4.49] | K=8: 2.45 [ 1.06, 4.25]
-- SCR (dB) for cumulative Pd=0.9 by K, Doppler random; gain vs CAlloc [95% day-cluster CI] --
IN1 K=1: 8.95 [ 5.56,11.69] | K=2: 10.00 [ 6.21,12.95] | K=4: 9.96 [ 5.96,13.07] | K=8: 9.45 [ 5.43,12.73]
NA4 K=1: 11.09 [ 7.22,14.10] | K=2: 12.33 [ 7.84,15.68] | K=4: 12.94 [ 8.38,16.44] | K=8: 13.26 [ 8.80,16.82]
IN4 K=1: 9.65 [ 5.85,12.64] | K=2: 10.99 [ 6.43,14.37] | K=4: 10.81 [ 6.09,14.37] | K=8: 10.15 [ 5.18,13.94]
G1 K=1: 9.01 [ 5.66,11.73] | K=2: 10.07 [ 6.39,13.05] | K=4: 10.05 [ 6.23,13.22] | K=8: 9.56 [ 5.56,12.83]
CA16 K=1: 1.41 [ 0.86, 2.14] | K=2: 1.43 [ 0.76, 2.32] | K=4: 1.20 [ 0.38, 2.32] | K=8: 0.53 [-0.47, 1.81]

=== per-look Pfa design 0.001 ===
clean cumulative false-alarm prob, K=1,2,4,8: IN1 0.0011 0.0017 0.0033 0.0063
clean cumulative false-alarm prob, K=1,2,4,8: G1 0.0013 0.0021 0.0040 0.0079
clean cumulative false-alarm prob, K=1,2,4,8: IN4 0.0008 0.0014 0.0027 0.0055
clean cumulative false-alarm prob, K=1,2,4,8: NA4 0.0007 0.0014 0.0027 0.0051
clean cumulative false-alarm prob, K=1,2,4,8: CA16 0.0007 0.0015 0.0025 0.0051
clean cumulative false-alarm prob, K=1,2,4,8: CAlloc 0.0010 0.0012 0.0018 0.0029
-- per-look Pd at SCR 10 dB, Doppler random (looks j=0..7) --
IN1 0.804 0.587 0.347 0.025 0.001 0.000 0.000 0.000
G1 0.833 0.626 0.455 0.014 0.002 0.000 0.000 0.000
IN4 0.802 0.559 0.133 0.010 0.000 0.000 0.000 0.000
NA4 0.852 0.790 0.516 0.312 0.173 0.124 0.114 0.110
CA16 0.555 0.368 0.125 0.025 0.006 0.002 0.001 0.000
CAlloc 0.290 0.290 0.289 0.290 0.289 0.291 0.289 0.289
-- per-look Pd at SCR 10 dB, Doppler matched (looks j=0..7) --
IN1 0.804 0.003 0.000 0.000 0.000 0.000 0.000 0.000
G1 0.833 0.008 0.002 0.001 0.000 0.000 0.000 0.000
IN4 0.802 0.002 0.000 0.000 0.000 0.000 0.000 0.000
NA4 0.852 0.448 0.023 0.001 0.000 0.000 0.000 0.000
CA16 0.555 0.364 0.066 0.004 0.001 0.000 0.000 0.000
CAlloc 0.290 0.289 0.291 0.290 0.289 0.289 0.290 0.290
-- per-look Pd at SCR 10 dB, Doppler opposite (looks j=0..7) --
IN1 0.804 0.878 0.629 0.057 0.001 0.000 0.000 0.000
G1 0.833 0.904 0.801 0.025 0.002 0.001 0.000 0.000
IN4 0.802 0.829 0.278 0.001 0.000 0.000 0.000 0.000
NA4 0.852 0.906 0.849 0.540 0.337 0.273 0.259 0.252
CA16 0.555 0.372 0.117 0.032 0.004 0.002 0.000 0.000
CAlloc 0.290 0.290 0.291 0.290 0.289 0.292 0.290 0.291
-- SCR (dB) for cumulative Pd=0.5 by K, Doppler random; gain vs CAlloc [95% day-cluster CI] --
IN1 K=1: 10.93 [ 7.28,14.02] | K=2: 11.99 [ 7.93,15.35] | K=4: 11.48 [ 7.26,15.09] | K=8: 10.88 [ 6.53,14.62]
NA4 K=1: 12.40 [ 9.09,15.50] | K=2: 13.19 [ 9.42,16.38] | K=4: 12.98 [ 9.27,15.90] | K=8: 12.73 [ 9.29,15.59]
IN4 K=1: 11.17 [ 7.43,14.42] | K=2: 12.12 [ 7.77,15.88] | K=4: 11.49 [ 7.08,15.34] | K=8: 10.87 [ 6.35,14.85]
G1 K=1: 11.63 [ 8.41,14.50] | K=2: 12.89 [ 9.11,16.16] | K=4: 12.54 [ 8.65,15.96] | K=8: 11.97 [ 8.03,15.55]
CA16 K=1: 4.06 [ 2.75, 5.93] | K=2: 4.00 [ 2.44, 6.10] | K=4: 3.57 [ 1.85, 5.85] | K=8: 3.00 [ 1.14, 5.43]
-- SCR (dB) for cumulative Pd=0.9 by K, Doppler random; gain vs CAlloc [95% day-cluster CI] --
IN1 K=1: 9.54 [ 5.98,12.63] | K=2: 10.64 [ 6.86,13.92] | K=4: 10.38 [ 6.25,13.85] | K=8: 9.80 [ 5.45,13.28]
NA4 K=1: 11.30 [ 7.70,14.40] | K=2: 12.63 [ 8.56,16.01] | K=4: 13.09 [ 9.01,16.52] | K=8: 13.06 [ 9.04,16.55]
IN4 K=1: 9.86 [ 5.87,13.40] | K=2: 11.02 [ 6.57,14.94] | K=4: 10.61 [ 5.84,14.63] | K=8: 10.00 [ 5.16,14.13]
G1 K=1: 10.17 [ 6.99,13.18] | K=2: 11.42 [ 7.90,14.55] | K=4: 11.26 [ 7.42,14.46] | K=8: 10.68 [ 6.87,14.03]
CA16 K=1: 2.45 [ 1.56, 3.81] | K=2: 2.39 [ 1.22, 4.07] | K=4: 2.00 [ 0.67, 3.77] | K=8: 1.40 [-0.11, 3.33]

```

Exploratory: comparison at equal dwell false-alarm probability (0.01), with max and sum-of-squares dwell statistics:

EXPLORATORY: units 56; dwell Pfa design 0.01; persistent Swerling-1 targets, random Doppler

```

-- K = 1 looks --
IN1 max: measured dwell Pfa 0.0099 | SCR for Pd 0.5 -0.9 dB | Pd 0.9 8.8 dB
IN1 sum: measured dwell Pfa 0.0099 | SCR for Pd 0.5 -0.9 dB | Pd 0.9 8.8 dB
NA4 max: measured dwell Pfa 0.0095 | SCR for Pd 0.5 -1.9 dB | Pd 0.9 6.6 dB
NA4 sum: measured dwell Pfa 0.0095 | SCR for Pd 0.5 -1.9 dB | Pd 0.9 6.6 dB
CAlloc max: measured dwell Pfa 0.0108 | SCR for Pd 0.5 9.0 dB | Pd 0.9 17.8 dB
CAlloc sum: measured dwell Pfa 0.0108 | SCR for Pd 0.5 9.0 dB | Pd 0.9 17.8 dB
gain of IN1-max over integrated power CFAR (CAlloc-sum) at Pd 0.5: 9.93 [ 6.46, 12.81] dB
gain of IN1-max over integrated power CFAR (CAlloc-sum) at Pd 0.9: 8.94 [ 5.54, 11.59] dB
gain of NA4-max over integrated power CFAR (CAlloc-sum) at Pd 0.5: 10.99 [ 7.58, 13.81] dB
gain of NA4-max over integrated power CFAR (CAlloc-sum) at Pd 0.9: 11.20 [ 7.46, 14.19] dB
gain of IN1-sum over integrated power CFAR (CAlloc-sum) at Pd 0.5: 9.93 [ 6.37, 12.81] dB
gain of IN1-sum over integrated power CFAR (CAlloc-sum) at Pd 0.9: 8.94 [ 5.54, 11.59] dB
gain of NA4-sum over integrated power CFAR (CAlloc-sum) at Pd 0.5: 10.99 [ 7.61, 13.81] dB
gain of NA4-sum over integrated power CFAR (CAlloc-sum) at Pd 0.9: 11.20 [ 7.46, 14.19] dB

-- K = 2 looks --
IN1 max: measured dwell Pfa 0.0102 | SCR for Pd 0.5 -2.0 dB | Pd 0.9 7.1 dB
IN1 sum: measured dwell Pfa 0.0097 | SCR for Pd 0.5 -2.2 dB | Pd 0.9 6.8 dB
NA4 max: measured dwell Pfa 0.0093 | SCR for Pd 0.5 -2.5 dB | Pd 0.9 4.8 dB
NA4 sum: measured dwell Pfa 0.0094 | SCR for Pd 0.5 -2.7 dB | Pd 0.9 4.1 dB
CAlloc max: measured dwell Pfa 0.0110 | SCR for Pd 0.5 8.1 dB | Pd 0.9 16.8 dB
CAlloc sum: measured dwell Pfa 0.0108 | SCR for Pd 0.5 8.7 dB | Pd 0.9 17.2 dB
gain of IN1-max over integrated power CFAR (CAlloc-sum) at Pd 0.5: 10.72 [ 6.85, 13.68] dB
gain of IN1-max over integrated power CFAR (CAlloc-sum) at Pd 0.9: 10.12 [ 6.29, 13.09] dB
gain of NA4-max over integrated power CFAR (CAlloc-sum) at Pd 0.5: 11.15 [ 7.71, 13.85] dB
gain of NA4-max over integrated power CFAR (CAlloc-sum) at Pd 0.9: 12.46 [ 7.90, 15.85] dB
gain of IN1-sum over integrated power CFAR (CAlloc-sum) at Pd 0.5: 10.91 [ 7.32, 13.74] dB
gain of IN1-sum over integrated power CFAR (CAlloc-sum) at Pd 0.9: 10.40 [ 6.73, 13.29] dB
gain of NA4-sum over integrated power CFAR (CAlloc-sum) at Pd 0.5: 11.42 [ 8.19, 13.92] dB
gain of NA4-sum over integrated power CFAR (CAlloc-sum) at Pd 0.9: 13.13 [ 8.71, 16.43] dB

-- K = 4 looks --
IN1 max: measured dwell Pfa 0.0094 | SCR for Pd 0.5 -1.9 dB | Pd 0.9 7.4 dB
IN1 sum: measured dwell Pfa 0.0099 | SCR for Pd 0.5 -2.6 dB | Pd 0.9 6.6 dB
NA4 max: measured dwell Pfa 0.0087 | SCR for Pd 0.5 -2.7 dB | Pd 0.9 4.1 dB
NA4 sum: measured dwell Pfa 0.0099 | SCR for Pd 0.5 -3.3 dB | Pd 0.9 2.7 dB
CAlloc max: measured dwell Pfa 0.0109 | SCR for Pd 0.5 7.6 dB | Pd 0.9 16.2 dB
CAlloc sum: measured dwell Pfa 0.0111 | SCR for Pd 0.5 8.3 dB | Pd 0.9 16.8 dB
gain of IN1-max over integrated power CFAR (CAlloc-sum) at Pd 0.5: 10.29 [ 6.26, 13.55] dB
gain of IN1-max over integrated power CFAR (CAlloc-sum) at Pd 0.9: 9.47 [ 5.33, 12.64] dB
gain of NA4-max over integrated power CFAR (CAlloc-sum) at Pd 0.5: 11.08 [ 7.57, 13.82] dB
gain of NA4-max over integrated power CFAR (CAlloc-sum) at Pd 0.9: 12.70 [ 7.98, 16.26] dB

```

```

gain of IN1-sum over integrated power CFAR (CAloc-sum) at Pd 0.5: 10.94 [ 7.48, 13.70] dB
gain of IN1-sum over integrated power CFAR (CAloc-sum) at Pd 0.9: 10.22 [ 6.66, 13.08] dB
gain of NA4-sum over integrated power CFAR (CAloc-sum) at Pd 0.5: 11.67 [ 8.74, 13.95] dB
gain of NA4-sum over integrated power CFAR (CAloc-sum) at Pd 0.9: 14.17 [ 9.70, 17.61] dB

-- K = 8 looks --
IN1  max: measured dwell Pfa 0.0093 | SCR for Pd 0.5 -1.3 dB | Pd 0.9 8.3 dB
IN1  sum: measured dwell Pfa 0.0102 | SCR for Pd 0.5 -2.7 dB | Pd 0.9 7.0 dB
NA4  max: measured dwell Pfa 0.0091 | SCR for Pd 0.5 -2.9 dB | Pd 0.9 4.3 dB
NA4  sum: measured dwell Pfa 0.0100 | SCR for Pd 0.5 -3.8 dB | Pd 0.9 2.4 dB
CAloc max: measured dwell Pfa 0.0103 | SCR for Pd 0.5 7.4 dB | Pd 0.9 16.0 dB
CAloc sum: measured dwell Pfa 0.0105 | SCR for Pd 0.5 8.0 dB | Pd 0.9 16.4 dB
gain of IN1-max over integrated power CFAR (CAloc-sum) at Pd 0.5: 9.31 [ 4.87, 12.82] dB
gain of IN1-max over integrated power CFAR (CAloc-sum) at Pd 0.9: 8.14 [ 3.83, 11.48] dB
gain of NA4-max over integrated power CFAR (CAloc-sum) at Pd 0.5: 10.85 [ 7.29, 13.54] dB
gain of NA4-max over integrated power CFAR (CAloc-sum) at Pd 0.9: 12.13 [ 7.50, 15.64] dB
gain of IN1-sum over integrated power CFAR (CAloc-sum) at Pd 0.5: 10.70 [ 7.35, 13.40] dB
gain of IN1-sum over integrated power CFAR (CAloc-sum) at Pd 0.9: 9.35 [ 6.02, 12.16] dB
gain of NA4-sum over integrated power CFAR (CAloc-sum) at Pd 0.5: 11.74 [ 9.17, 13.75] dB
gain of NA4-sum over integrated power CFAR (CAloc-sum) at Pd 0.9: 14.01 [ 9.97, 17.21] dB

```

Exploratory: exact-law prediction of the post-onset decay:

```

Exact-law prediction vs measured per-look Pd after onset (IN-ARCP, SCR 10 dB, Pfa 0.01); pooled over 56 units (exploratory)
random  predicted 0.885 0.710 0.635 0.472 0.086 0.012 0.004 0.002
        measured 0.885 0.711 0.639 0.478 0.084 0.012 0.005 0.002   MAE per unit x look 0.010
matched  predicted 0.885 0.023 0.013 0.007 0.004 0.002 0.002 0.001
        measured 0.885 0.024 0.013 0.007 0.003 0.002 0.002 0.001   MAE per unit x look 0.003
opposite predicted 0.885 0.941 0.906 0.793 0.167 0.014 0.004 0.002
        measured 0.885 0.940 0.905 0.796 0.165 0.014 0.004 0.002   MAE per unit x look 0.008

```

8 Numerical checks of the detection theory

Monte Carlo checks of Corollary 2, Proposition 2 (closed forms) and Proposition 3 (output of `detection/theory_checks.py`):

```

T1/T2 exact detection law vs Monte Carlo (Swirling-1 target in Y only; 400k draws)
rho=0.91-0.18j c=None r=0.91-0.18j SCR=1.0: MC 0.5358 exact 0.5364 closed form 0.5364
rho=0.91-0.18j c=10.0 r=0.88-0.18j SCR=3.0: MC 0.6381 exact 0.6374
rho=0.70 c=1.0 r=0.65 SCR=10.0: MC 0.4275 exact 0.4263

T3 noise-limited whitening gain: empirical (c+1)/E|Y-pred|^2 vs closed forms (m=16 history, 200k draws)
|rho|=0.7 c= 0.1: IN emp -1.46 dB closed -1.46 | oracle emp 0.03 dB closed(m=inf) 0.03 | clutter-only bound 2.92
|rho|=0.7 c= 1.0: IN emp -0.01 dB closed 0.00 | oracle emp 0.66 dB closed(m=inf) 0.67 | clutter-only bound 2.92
|rho|=0.7 c= 10.0: IN emp 2.24 dB closed 2.23 | oracle emp 2.29 dB closed(m=inf) 2.27 | clutter-only bound 2.92
|rho|=0.7 c= 100.0: IN emp 2.85 dB closed 2.84 | oracle emp 2.85 dB closed(m=inf) 2.84 | clutter-only bound 2.92
|rho|=0.93 c= 0.1: IN emp -2.32 dB closed -2.32 | oracle emp 0.12 dB closed(m=inf) 0.12 | clutter-only bound 8.69
|rho|=0.93 c= 1.0: IN emp 0.02 dB closed 0.00 | oracle emp 1.66 dB closed(m=inf) 1.65 | clutter-only bound 8.69
|rho|=0.93 c= 10.0: IN emp 5.34 dB closed 5.34 | oracle emp 5.76 dB closed(m=inf) 5.76 | clutter-only bound 8.69
|rho|=0.93 c= 100.0: IN emp 8.17 dB closed 8.18 | oracle emp 8.18 dB closed(m=inf) 8.19 | clutter-only bound 8.69
|rho|=0.98 c= 0.1: IN emp -2.50 dB closed -2.52 | oracle emp 0.20 dB closed(m=inf) 0.21 | clutter-only bound 14.02
|rho|=0.98 c= 1.0: IN emp -0.01 dB closed 0.00 | oracle emp 2.22 dB closed(m=inf) 2.22 | clutter-only bound 14.02
|rho|=0.98 c= 10.0: IN emp 6.69 dB closed 6.69 | oracle emp 7.78 dB closed(m=inf) 7.78 | clutter-only bound 14.02
|rho|=0.98 c= 100.0: IN emp 12.31 dB closed 12.32 | oracle emp 12.43 dB closed(m=inf) 12.44 | clutter-only bound 14.02

T4 persistent strong target: limiting score and Pd (IN-ARCP, r fitted = 0.93e^{-0.2j})
sqrt(m/(m-1)) = 1.033; thresholds q(0.1)=1.574 q(0.01)=2.310
Doppler w=+0.0: limiting score 0.937 | SCR 0 dB: Pd(0.01) 0.010 | SCR 20 dB: Pd(0.01) 0.000 | SCR 40 dB: Pd(0.01) 0.000
Doppler w=-0.2: limiting score 0.613 | SCR 0 dB: Pd(0.01) 0.010 | SCR 20 dB: Pd(0.01) 0.000 | SCR 40 dB: Pd(0.01) 0.000
Doppler w=+1.0: limiting score 1.029 | SCR 0 dB: Pd(0.01) 0.000 | SCR 20 dB: Pd(0.01) 0.000 | SCR 40 dB: Pd(0.01) 0.000

```

9 Baselines: parametric texture and adaptive conformal inference

B1/B2 within-session units: 56; transfer days: 6

```

=== 1-alpha = 0.9 ===
B1 NA4-K / NA4 radius (GM): 1.0092 [1.0064, 1.0115] | coverage NA4 0.9019 NA4-K 0.9015 | max dev NA4 0.043 NA4-K 0.043
B2 within ACI g=0.005: coverage 0.9010 vs split 0.9013 | radius ratio (finite) 0.999 [0.997,1.000] | infinite-region fraction 0.0000 | max dev 0.033 vs 0.036
B2 within ACI g=0.02: coverage 0.9007 vs split 0.9013 | radius ratio (finite) 0.999 [0.995,1.002] | infinite-region fraction 0.0000 | max dev 0.029 vs 0.036
B2 transfer IN1 : mean coverage 0.9029 worst day 0.8822 | radius vs split 1.000 | max dev 0.024 | per day 0.905 0.894 0.915 0.918 0.904 0.882
B2 transfer ACIO.005 : mean coverage 0.9032 worst day 0.8889 | radius vs split 1.001 | max dev 0.020 | per day 0.904 0.897 0.912 0.914 0.903 0.889
B2 transfer ACIO.02 : mean coverage 0.9024 worst day 0.8965 | radius vs split 1.000 | max dev 0.015 | per day 0.903 0.900 0.906 0.908 0.902 0.897

=== 1-alpha = 0.99 ===
B1 NA4-K / NA4 radius (GM): 1.0078 [1.0034, 1.0104] | coverage NA4 0.9905 NA4-K 0.9906 | max dev NA4 0.011 NA4-K 0.011
B2 within ACI g=0.005: coverage 0.9903 vs split 0.9901 | radius ratio (finite) 1.001 [0.996,1.007] | infinite-region fraction 0.0000 | max dev 0.010 vs 0.011
B2 within ACI g=0.02: coverage 0.9901 vs split 0.9901 | radius ratio (finite) 1.006 [0.998,1.015] | infinite-region fraction 0.0115 | max dev 0.009 vs 0.011
B2 transfer IN1 : mean coverage 0.9921 worst day 0.9875 | radius vs split 1.000 | max dev 0.006 | per day 0.987 0.992 0.994 0.996 0.994 0.989
B2 transfer ACIO.005 : mean coverage 0.9918 worst day 0.9889 | radius vs split 0.997 | max dev 0.005 | per day 0.989 0.991 0.993 0.995 0.993 0.990
B2 transfer ACIO.02 : mean coverage 0.9912 worst day 0.9902 | radius vs split 0.995 | max dev 0.004 | per day 0.990 0.990 0.992 0.993 0.991 0.990

```

10 Second radar: full results

units 24 (station x Rx x rotation); test episodes per CNR class {'<5': 721095, '5..0': 265416, '0..5': 229845, '5..10': 68556, '10..20': 31870, '>=20': 338}
mean CNR per class (dB): -12.9 -2.2 2.4 7.2 13.4 20.1

```

=== 1-alpha = 0.9 ===
J1 marginal coverage: IN1 0.9000 | IN4 0.9000 | NA1 0.8999 | NA4 0.9000 | G1 0.8974 | NA4G 0.8933 | U 0.8998 | RMS 0.9000 | CA16 0.9000 | CAloc 0.9001
J2 coverage by CNR class (mean over units):
IN1 0.897 0.897 0.900 0.917 0.962 1.000 | mean max class deviation 0.066
NA1 0.939 0.850 0.840 0.863 0.882 0.945 | mean max class deviation 0.067
NA4 0.917 0.864 0.881 0.912 0.946 1.000 | mean max class deviation 0.055
U 0.972 0.896 0.824 0.678 0.305 0.017 | mean max class deviation 0.629
CA16 0.893 0.895 0.904 0.942 0.985 1.000 | mean max class deviation 0.086
IN1 exact law 0.896 0.900 0.905 0.914 0.932 0.966 | MAE per unit x class 0.0104

=== 1-alpha = 0.99 ===
J1 marginal coverage: IN1 0.9900 | IN4 0.9900 | NA1 0.9900 | NA4 0.9901 | G1 0.9897 | NA4G 0.9846 | U 0.9899 | RMS 0.9900 | CA16 0.9900 | CAloc 0.9900
J2 coverage by CNR class (mean over units):
IN1 0.989 0.989 0.990 0.995 0.999 1.000 | mean max class deviation 0.009
NA1 0.996 0.983 0.979 0.984 0.989 1.000 | mean max class deviation 0.013
NA4 0.993 0.983 0.987 0.993 0.998 1.000 | mean max class deviation 0.009
U 1.000 0.997 0.991 0.959 0.734 0.245 | mean max class deviation 0.305

```

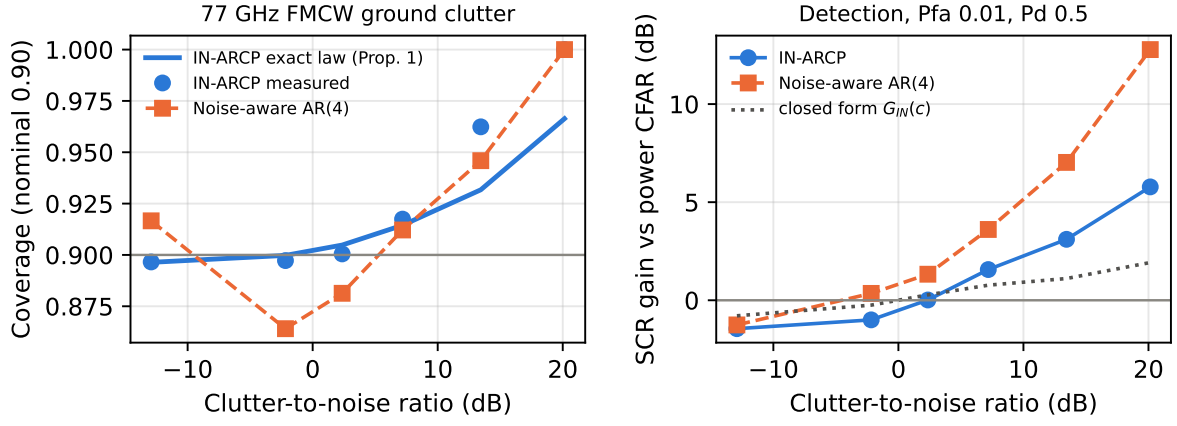


Figure 5: 77 GHz FMCW ground clutter: coverage by CNR class with the exact law, and SCR gain over the local power detector.

```

CA16  0.989  0.989  0.992  0.998  1.000  1.000 | mean max class deviation 0.010
IN1 exact law  0.989  0.990  0.991  0.993  0.995  0.998 | MAE per unit x class 0.0019

J3 detection (Swirling 1, Pfa 0.01): SCR gain vs CAlloc at Pd=0.5 per CNR class, mean [95% CI]; closed form  $G_{IN}$ 
 $G_{IN}(c)$  dB      -0.79      -0.25      0.28      0.77      1.11      1.92
IN1      -1.45 [-1.79,-1.13] -1.00 [-1.19,-0.83] 0.02 [-0.23, 0.23] 1.57 [ 0.89, 2.20] 3.11 [ 1.97, 4.26] 5.78 [ 5.78, 5.78]
IN4      -1.42 [-1.57,-1.28] -0.85 [-0.99,-0.72] 0.52 [ 0.03, 0.96] 3.19 [ 2.27, 3.98] 6.78 [ 5.08, 8.34] 12.33 [12.33,12.33]
NA1      -2.10 [-2.65,-1.61] 0.65 [ 0.31, 0.93] 1.99 [ 1.06, 2.70] 4.21 [ 2.71, 5.51] 7.81 [ 5.26,10.08] 13.07 [13.07,13.07]
NA4      -1.26 [-1.38,-1.14] 0.35 [ 0.08, 0.64] 1.33 [ 0.78, 1.88] 3.61 [ 2.70, 4.44] 7.03 [ 5.50, 8.59] 12.78 [12.78,12.78]
U        -5.48 [-6.02,-5.02] -2.19 [-2.65,-1.80] 0.22 [-0.21, 0.61] 4.11 [ 3.54, 4.65] 11.44 [10.93,11.95] 13.07 [13.07,13.07]
CA16     -0.62 [-0.64,-0.59] -0.49 [-0.55,-0.44] -0.47 [-0.50,-0.46] -0.54 [-0.57,-0.51] -0.61 [-0.67,-0.54] -0.85 [-0.85,-0.85]

J4 interference scenario C (Y-chirp hit rate 0.020; hit episodes 1920, clean 53520)
IN1  coverage 0.90: clean 0.901 hit 0.792 | 0.99: clean 0.990 hit 0.979
NA1  coverage 0.90: clean 0.828 hit 0.584 | 0.99: clean 0.964 hit 0.854
NA4  coverage 0.90: clean 0.870 hit 0.644 | 0.99: clean 0.980 hit 0.926
U    coverage 0.90: clean 0.761 hit 0.449 | 0.99: clean 0.950 hit 0.785
CA16 coverage 0.90: clean 0.900 hit 0.719 | 0.99: clean 0.989 hit 0.965

```